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Spectrally selective approaches for passive cooling of solar cells: A review

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HIGHLIGHTS

- The principle of spectrally selective approaches for cooling solar cells is analyzed.
- Radiative cooling is a significant method for solar cells to dissipate waste heat.
- Full-spectrum thermal management of solar cells is a novel way to cool solar cells passively.
- The cooling effect of full-spectrum thermal management is better than radiative cooling.
- Appropriate material is the biggest challenge for this new cooling technology.

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ABSTRACT

Waste heat is generated in solar cells during the daytime because they operate with a limited electrical efficiency. Moreover, the generated heat increases the solar cells' operating temperature, and this has an adverse effect on their electrical efficiency. Therefore, numerous cooling methods have been developed to cool solar cells, such as forced air/water flow, hybrid photovoltaic/thermal system, and phase change material based photovoltaic application. Recently, a novel concept of changing the solar cells' spectral response to both sunlight and thermal radiation has been proposed and developed to provide a passive cooling method for solar cells, which has drawn much attention from materials science to engineering fields. In this paper, the recent advancements of such a spectrally selective approach to passively cool solar cells, including radiative cooling of solar cells and full-spectrum thermal management of solar cells are reviewed, analyzed, and discussed from fundamental principles to detailed demonstration. Furthermore, the technical challenges involved in developing this new cooling technology are discussed. This paper is devoted to give a systematic introduction to a new passive cooling technology for solar cells by controlling the spectral property of solar cells, aiming to open a new opportunity for solar cells' cooling.

1. Introduction

Photovoltaic (PV) conversion through solar cells is one of the great technologies for humans because it offers the opportunity to covert incident photons directly into electricity in a pollution free and simple way. Thus, solar cells are the core components of the PV technology [1]. There are numerous types of solar cells for PV conversions, such as silicon-based solar cells [2], gallium arsenide (GaAs) solar cells [3], copper indium gallium selenide (CIGS) solar cells [4], and perovskite solar cells [5]. At present, silicon-based solar cells are comparatively the excellent and popular option for PV conversion at an industrial scale because of their low production cost. However, the electrical efficiency of this type of solar cell is limited and only a part of absorbed solar energy can be used to generate electricity. The remaining absorbed solar power is dissipated into heat and increases the operating temperature of solar cells, and this is undortunately a major issure because the electrical efficiency is generally lower at elevated operating temperature. For example, a 1 °C increase in the operating temperature decreases the relative electrical efficiency of the crystalline silicon solar cells by approximately 0.4–0.5% [6], according to experiment results. Therefore, minimizing the operating temperature of solar cells is significant for PV applications.

In previous studies and applications, a variety of cooling technologies have been developed and used to cool solar cells. One of the most popular cooling methods is to enhance the convective heat transfer process between solar cells and the local ambient environment by additional media (e.g., ambient air [7] and water [8]), and this method can be divided into two different categories. The first begins by

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transferring the heat into the media and thereafter dissipating the collected heat from media into the ambient environment. Implementations that adopt this method include forced air flow [9], water spraying [10], water immersing [11], and heat pipe cooling [12]. For example, Teo et al. [7] designed a parallel array of ducts coupled with air flow for active cooling of solar cells, which demonstrated up to nearly 3–4% improvements in electrical efficiency over when active cooling was not applied. The second category involves harnessing the waste heat from solar cells for second utilization, which is commonly known as a photovoltaic/thermal system [13] For instance, Yu et al. [14] proposed a novel Micro-channel loop-heat-pipe structure to simultaneously cool solar cells and collect thermal energy. An electrical efficiency of nearly 19% and thermal collection efficiency ranging from 25.2% to 62.6% were measured during their testing period, which is claimed to be 17.2% higher than that of the existing photovoltaic/thermal hybrid system. Indeed, including additional cooling media and forced flow are very effective ways to promote heat dissipation from solar cells. However, these cooling methods require a power consumption (e.g., electricity), and the requirement for additional components on each panel greatly limits their feasibility for the industrial-scale applications. Aside from the above-mentioned cooling methods, using phase change material (PCM) is another common approach to cool solar cells [15]. For example, Lu et al. [16] integrated the PCM into a concentrating photovoltaic system to cool solar cells, which reduced the average temperature of solar cells by up to 20–25 °C and accordingly increase the electrical efficiency by 10–12%. However, one of the main challenges behind the PCM based PV cooling technology is that a matched PCM is urgently required. This PCM needs to exhibit excellent performances in terms of thermal conductivity, energy storage density, chemical and physical stability and so on [17]. Moreover, adopting the thermoelectric (TE) devices as a thermoelectric cooler (TEC) [18] or as a thermoelectric generator (TEG) [19] in the PV/TE system is another recent concept to cool solar cells. The TE device is advantageous in that it is silent, compact and does not require moving parts, thus it is reliable and durable [20]. However, state of the art TE devices are currently still impractical to cool solar cels; As a TEC, the power consumption required to cool solar cells is even higher than the gain in electricity from reducing its temperature, and as a TEG, it introduces a thermal resistance that would conversely increase the solar cells' temperature and negate the benefit of electricity gain from then TEG. Notably, plenty of review papers on the topic of general cooling technologies for solar cells have been published [21–23], which contains the detailed information of the aforementioned cooling approaches.

In recent years, the novel concept of changing the spectral response of solar cells to electromagnetic waves for achieving the passive cooling of solar cells has been proposed and developed. The fundamental principle of this spectrally selective method is to enhance the thermal emission of the solar cells [24] and/or decrease the solar cells' absorption of waste photons that are unable to cause electron-hole pair generation [25]. Unlike the aforementioned traditional cooling solutions, this spectrally selective approach can decrease the operating temperature of solar cells passively (i.e., no extra enery input required), and moreover, no moving parts nor any additional external components are required during the cooling process, thus making it greatly advantageous for industrial-scale implementation. At present, this topic has drawn much attention from basic material sciences to engineering fields. Motivated by this innovative concept and its potential, the developments and advancements on this topic are compiled and discussed in this paper. In doing so, a systematic introduction to a new passive cooling technology for solar cells by controlling the spectral property of solar cells is conducted, which attempts to open a new opportunity for solar cells' cooling.

In this review article, the energy balance of the solar cells and fundamental principles of the spectrally selective method for passive cooling of solar cells are introduced and analyzed in Section 2. A detailed discussion about these spectrally selective approaches that include radiative cooling of solar cells and full-spectrum thermal management of solar cells are then presented in Section 3. Section 4 emphasizes the challenges and prospects of this new technology, which give some new topics for further investigations.

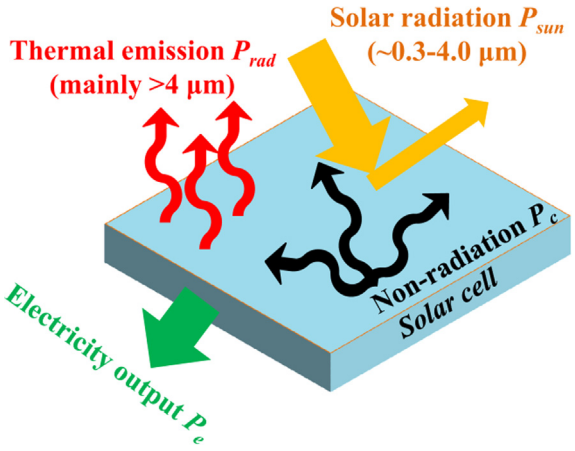


Fig. 1. Schematic of the energy balance of a solar cell.

2. Fundamental principles

2.1. Energy balance of solar cells

Consider a solar cell with a temperature of T_{sc} exposed to the sun and sky. Four main energy fluxes flow into and out of the solar cell (see Fig. 1), which are the thermal emission of the solar cell P_{rad}, solar radiation absorbed by the solar cell P_{sun}, non-radiative heat dissipation P_c, and electricity output P_e. All these terms strongly influence the operating temperature of the solar cell.

According to the first law of thermodynamics, the steady-state energy equation of the solar cell is as follows:

$$P_{sun} - P_{rad} - P_c - P_e = 0 \quad (1)$$

The absorbed solar radiation is calculated by:

$$P_{sun} = G_{0.3-\lambda_{eff}}\alpha_{0.3-\lambda_{eff}} + G_{\lambda_{eff}-4}\alpha_{\lambda_{eff}-4} \quad (2)$$

where λ_{eff} denotes the effective wavelength of the solar cell (as shown in Fig. 2), which means that photons with wavelengths below λ_{eff} can cause electron-hole pair of the solar cell and be partially converted into electricity, whereas the other photons cannot generate electricity. For example, for crystalline silicon solar cells, the λ_{eff} is approximately

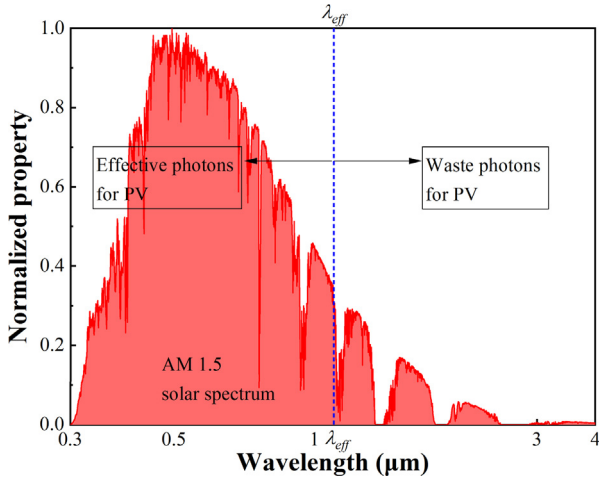


Fig. 2. Schematic of effective wavelength for PV conversion with normalized AM 1.5 solar spectrum plotted for a Ref. [26].

1.1 μm due to its natural energy bandgap. G denotes solar radiation power, and α represents the solar absorptivity of the solar cell.

The thermal emission P_{rad} of the solar cell is the net radiative power that combines the thermal radiative power of the solar cell and atmospheric radiation absorbed by the solar cell. Generally, P_{rad} can be calculated as [27]:

$$P_{rad} = \varepsilon_{sc}\sigma(T_{sc}^4 - T_{sky}^4) \tag{3}$$

where ε_{sc} denotes the thermal emissivity of the solar cell; σ is the Stefan-Boltzmann constant; T_{sc} and T_{sky} represent the temperature of solar cells and effective sky temperature. Notably, there are many correlations to describe the effective sky temperature [28], which are closely related to the local meteorological parameters.

The non-radiative heat dissipation of the solar cell P_c that contains the effect of conduction and convection is given by:

$$P_c = U(T_{sc} - T_{le}) \tag{4}$$

where T_{le} is the temperature of the local environment and U denotes the effective heat transfer coefficient between the solar cell and local environment.

The electricity output of the solar cell P_e is the only subject of interest during PV conversion. Generally, the value of P_e is affected by numerous factors, including the effective solar absorption of solar cells [29] and the operating temperature of solar cells [6]. In this paper, improving P_e by decreasing the operating temperature of solar cells is the main topic to be discussed.

2.2. The methods to passively reduce heat stack in solar cells

In this subsection, we analyze the possibility to cool solar cells from different aspects. Combining Eqs. (1)-(4), the electricity output P_e can be calculated by:

$$\begin{aligned} P_e &= \underbrace{G_{0.3-\lambda_{eff}}\alpha_{0.3-\lambda_{eff}}}_{I} + \underbrace{G_{\lambda_{eff}-4}\alpha_{\lambda_{eff}-4}}_{II} - \underbrace{\varepsilon_{sc}}_{III}\sigma(T_{sc}^4 - T_{sky}^4) \\ &\quad - \underbrace{U}_{IV}(T_{sc} - T_{le}) \end{aligned} \tag{5}$$

For solar cells (especially for semiconductor material based solar cells), the electricity output is mainly determined by the solar cells' effective solar absorption $\alpha_{0.3\text{-}\lambda eff}$ (i.e., item I in Eq. (5)) and intrinsic electrical efficiency. The intrinsic electrical efficiency of solar cells is determined by their materials, which is not the point discussed here. To ensure that the electricity output of solar cells is at a high level, the effective solar absorption $\alpha_{0.3\text{-}\lambda eff}$ should be high enough. At present, various methods, such as light trapping technology [30], have been used to improve the effective solar absorption of solar cells. Thus, for most solar cell types, including commercial silicon-based solar cells, the effective solar absorption $\alpha_{0.3\text{-}\lambda eff}$ is high enough to harvest photons. Based on the above analysis, it is known that the absorption of waste photons that cannot generate electricity is one of the main heat sources of solar cells. Thus, decreasing the absorption of waste photons $\alpha_{\lambda eff\text{-}4}$ (i.e., item II in Eq. (5)) is an efficient way to control the operating temperature of solar cells from the heat source point of view.

Apart from decreasing the waste-heat generation, enhancing the heat dissipation of solar cells is another effective method to cool solar cells. According to Eq. (5), there are two kinds of thermal dissipation; The first is thermal emission and the other is convective heat transfer. In the former case, increasing the thermal emissivity ε_{sc} (i.e., item III in Eq. (5)) of the solar cells is the direct solution and this approach is closely related to the optical property of solar cells. In the second method, improving the value of the effective heat transfer coefficient U (i.e., item IV in Eq. (5)) is the primary focus and it is the most commonly used cooling technologies for solar cells, including forced air/water flow and hybrid PV/T systems.

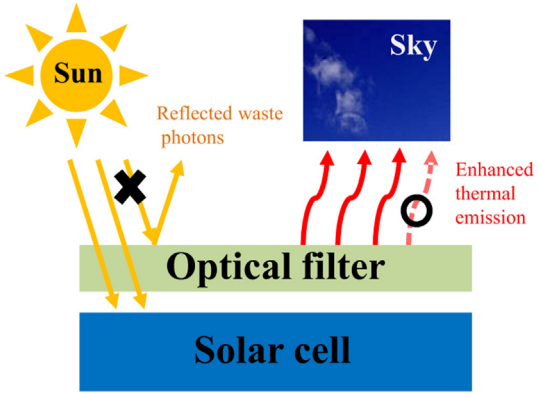


Fig. 3. Schematic of passive cooling of solar cells. An optical filter was put on top of the solar cell to change the spectral response of the solar cell.

2.3. Spectrally selective concept for passive cooling of solar cells

The general solution to passively cool solar cells on the basis of their spectral property is by adding a spectrally selective filter on the top of solar cells, as shown in Fig. 3.

The ideal optical filter needs to meet the following spectral requirements:

(1) For solar radiation below the effective wavelength λ_{eff} of solar cells, the filter needs to be highly transparent to ensure that photons with high energy can reach the solar cell and be partly converted into electricity;
(2) For solar radiation above the effective wavelength λ_{eff} of solar cells, the filter needs to have high reflectivity to reflect waste photons to decrease the waste-heat generation of solar cells;
(3) For the thermal radiation band, the filter needs to exhibit strong thermal emission to maximally dissipate the solar cells' waste heat into the cold sky.

Notably, the above three pieces of spectral requirements are proposed for the ideal filter. At present, there are two typical kinds of filters that can be used to cool solar cells passively. The first kind of filter meets the above (1) and (3) spectral requirements and this kind of cooling technology is called ***radiative cooling of solar cells***. The second kind of filter is close to the ideal filter, which meets all the spectral requirements proposed above, and this cooling technology can be named ***full-spectrum thermal management of solar cells***. In section 3, these two spectrally selective cooling approaches for solar cells will be reviewed, compiled, and discussed.

3. Spectrally selective approaches

In this section, the works of recent literature that use spectrally selective approaches to cool solar cells are compiled and discussed. The cooling performance of different spectrally selective approaches and the related optical materials are analyzed and compared.

3.1. Radiative cooling

Radiative cooling of solar cells is a hot topic in the early stage of its research. The main principle behind this approach is to add a transparent cooler on top of solar cells to enhance the thermal emissivity of solar cells and simultaneously maintain the effective solar absorption of solar cells. Thus, the transparent cooler needs to have high transmittance in the solar radiation wavelength band and exhibit strong thermal emission in the mid-infrared wavelength band.

Zhu et al. systematically proposed and demonstrated the concept of radiative cooling of solar cells. First, the operating temperature of bare

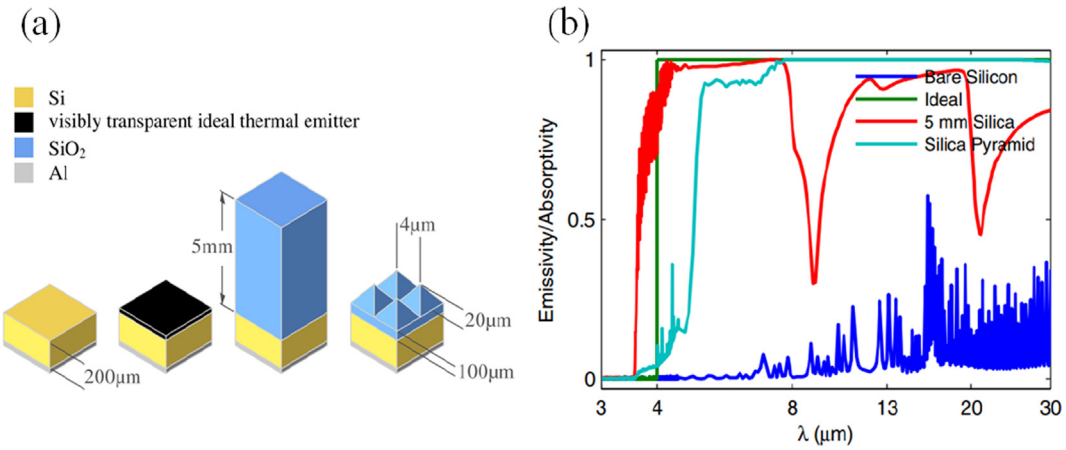


Fig. 4. (a) Schematic of a pyramid silica cooler design and (b) the corresponding spectral emissivity over the mid-infrared wavelength band. The figures are taken from Ref. [24].

crystalline silicon solar cells was predicted to be potentially reduced by over 18 °C by using an ideal cooler [24], which in turn could improve the relative electrical efficiency by 7.2–9.0%. Then, a transparent cooler that consists of a 2D square lattice of silica pyramids and a 100-μm-thick uniform silica layer was designed, as shown in Fig. 4a. This cooler is optically transparent and emits as a blackbody in the mid-infrared wavelength band. Performance prediction results showed that, even at 800 $W \cdot m^{-2}$ solar heating power, the temperature reduction of silicon solar cells with this pyramid cooler desingn could reach 17.6 K, which subsequently improves the relative electrical efficiency by 7.9%. Furthermore, a similar polydimethylsiloxane (PDMS) pyramid cooler was also proposed and investigated by Lee et al. [31]. A thermal analysis showed that a pyramid-structured PDMS layer can significantly reduce the temperature of an organic, a perovskite, and a micro-crystalline flexible solar cell by approximately 11–16 °C.

Apart from the conceptual development and theoretical analysis of radiative cooling of solar cells, Zhu et al. [32] designed and fabricated a transparent cooler (as shown in Fig. 5a) by etching grooves of air holes deep into a 500-μm-thick double-side–polished fused silica wafer. The spectral characterization revealed that such a groove cooler has an excellent thermal emissivity in the mid-infrared wavelength band and has high solar transmittance (as shown in Fig. 5b). To emphasize the effect of radiative cooling, comparative experiments (as shown in Fig. 5c) were conducted under the condition of isolating the effects of ambient wind on the solar cells and the results showed that the temperature of silicon solar cells can be passively reduced by 13 °C, which is close to the predicted results reported in Ref. [24]. However, as the solar cells or solar panels are often exposed to ambient air directly, the authors repeated the experiment that includes the effect of natural convection and showed experimentally that the temperature of solar cells could be reduced by up to 5.2 °C, which is still a useful temperature reduction for photovoltaic applications.

Based on the above several pieces of research, it is found that the radiative cooling method has a significant and efficient effect on cooling solar cells, which opens up a new degree of freedom for the cooling of solar cells using passive methods. However, there are still

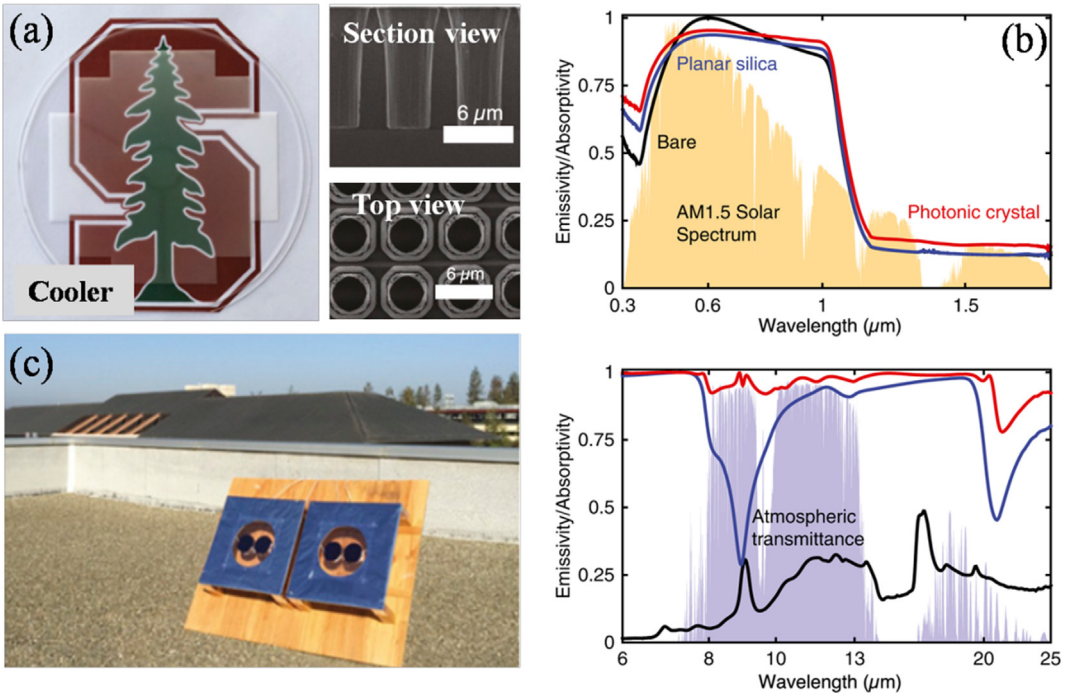


Fig. 5. (a) Schematic of an air holes grooves silica cooler design. (b) Measured spectral property of the designed cooler. (c) The outdoor experiment apparatus. The figures and data are taken from Ref. [32].

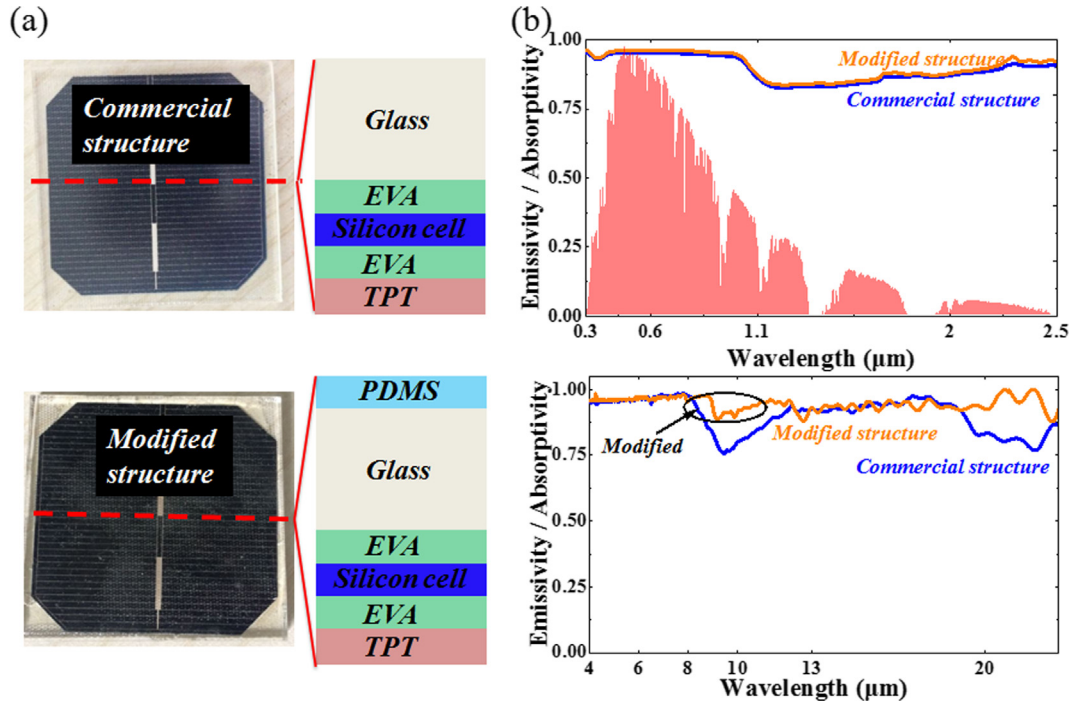


Fig. 6. (a) Schematic of encapsulated silicon solar cell with a glass and a modified structure, where the latter has an additional PDMS layer to enhance the thermal emissivity. (b) Measured spectral property of the two samples. The figures and data are taken from Ref. [36].

two concerns that need to be clarified. First, the bare solar cells used in the above studies are simplified silicon solar cells whose intrinsic thermal emissivity in the mid-infrared wavelength band is relatively low. Second, solar cells are generally covered with encapsulating materials (e.g., glass [33] and polyester materials [34]) whose thermal emissivity at ambient temperature can be already over 0.8 [35]. Based on the above two concerns, the temperature reduction of solar cells by radiative cooling in the real PV conversion process may be an over-prediction.

To estimate the actual cooling performance of radiative cooling for encapsulated solar cells, Zhao et al. [36] conducted an outdoor experiment. A transparent glass was applied to encapsulate the silicon solar cell (commercial structure, as shown in Fig. 6a), which is the commonly used structure for PV panels. Besides, a thin PDMS layer was used to enhance the thermal emissivity of the encapsulated silicon solar cell (modified structure, as shown in Fig. 6a). The outdoor experiment results showed that the temperature of the two structures is consistent during the entire testing period, indicating that enhancing the radiative cooling capacity on the basis of glass encapsulated solar cells is limited. However, some contributions are still devoted to enhancing the performance of the radiative cooling of solar cells by modifying the radiative property of the glass cover. For example, Zhou et al. theoretically improved the thermal emissivity of low-iron soda-lime glass by an additional multilayer [37] or patterned surface [38] to enhance the radiative cooling capacity of the glass cover. Furthermore, Gentle and Smith [39] proposed a new idea to cool encapsulated solar cells by considering the angular property of the encapsulated materials (e.g., glass), and a further 1.5 °C temperature reduction of encapsulated solar cells is possible.

Another interesting finding in this topic is that when the designed coolers are placed on the top of bare solar cells, the solar absorption of solar cells can be enhanced in addition to the already improved radiative cooling capacity. The main reason for this phenomenon is that some coolers passively act as the anti-reflection layer and/or light-trapping structure. Lu et al. [40] designed and fabricated a silica texture cooler (as shown in Fig. 7a) for bare solar cells. The optical test demonstrated that the silica texture exhibits a higher transmittance (as shown in Fig. 7b) within the visible-Near infrared wavelength band than commonly used glass, which will improve the effective optical efficiency of solar cells and correspondingly improve the electrical efficiency. Besides, the thermal emission of the designed silica texture is stronger than bare glass, indicating that the radiative cooling capacity of the designed cooler is better than bare glass. Eventually, the electrical test proved that the electrical efficiency of the solar cell with a designed silica texture cooler is higher than that with bare glass, with a relative improvement of 3.13%. This improvement is due to the contributing collection of sunlight and the thermal management of solar cells. Also, a similar finding was obtained by Long et al. [41] by putting a micro-grating silica coating on top of silicon solar cells.

Although the aforementioned kinds of solar cells and PV applications are silicon-based solar cells and flat plate solar PV conversion (i.e., no solar concentration), radiative cooling methods can be also used to cool other types of solar cells (e.g., CIGS [42] and GaAs [43] solar cells) and be used in other PV applications (e.g., solar concentrated PV (CPV) [44], thermal PV (TPV) [38], and outer space PV applications [45]). For example, Zhou et al. [44] demonstrated an enhanced radiative cooling for low-bandgap PV cells under concentrated sunlight, as shown in Fig. 8. Experimental results showed that the operating temperature of solar cells was passively reduced by 10 °C, corresponding to a relatively open-circuit voltage improvement of 5.7% and an estimated lifetime increase of approximately 40% under 13 concentrated suns.

3.2. Full-spectrum thermal management of solar cells

The main feature of the radiative cooling of solar cells is that solar transparent coolers are used to enhance the thermal emission of solar cells and simultaneously maintain the solar absorption of solar cells. However, considering the fact that the commercial solar cells and encapsulated materials (e.g., glass) already have a high thermal emissivity, further improvement to the actual cooling effect of radiative cooling is limited. Li et al. [25] first proposed a full-spectrum thermal management method to cool solar cells by simultaneously enhancing radiative cooling while also selectively utilizing solar radiation. The schematic of this cooling approach and the ideal spectral property of

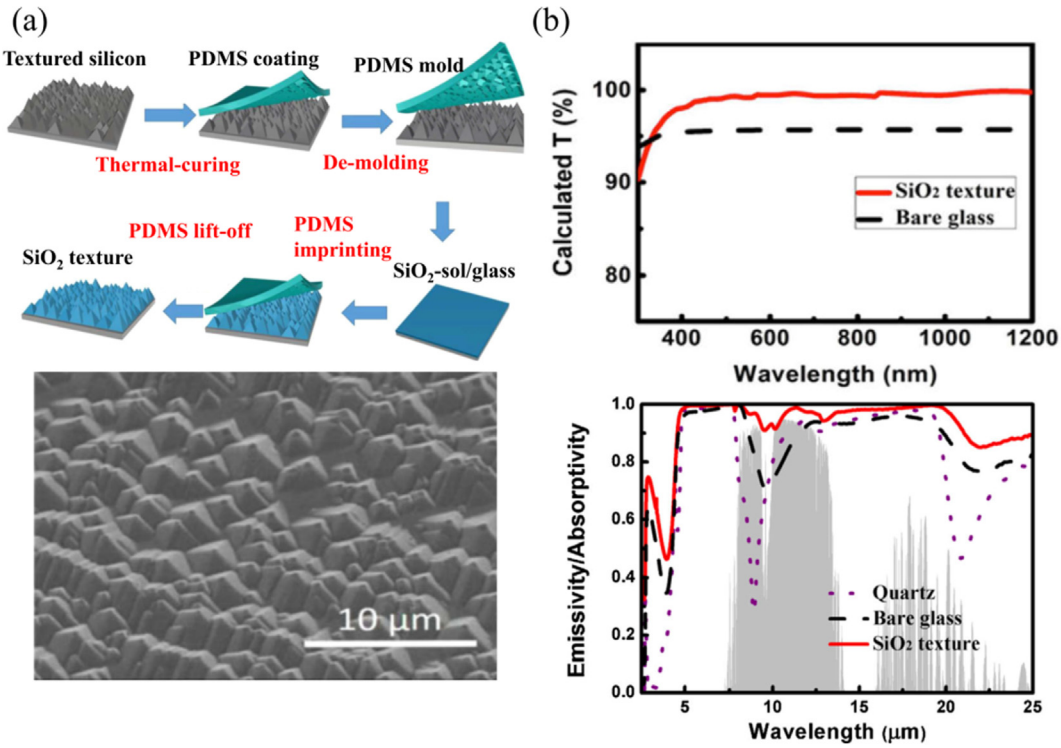


Fig. 7. (a) The fabrication process of silica texture and corresponding scanning electron microscope (SEM) image of the fabricated silica texture. (b) The spectral transmittance and emissivity of the fabricated silica texture. The figures and data are taken from Ref. [40].

the cooler are presented in Fig. 9. Here, a commercial silicon solar cell was selected as the primary device whose effective wavelength λ_{eff} was determined as 1.1 μm due to its intrinsic bandgap energy. A multilayer photonic cooler was designed using materials of silica, titanium dioxide, silicon nitride, and aluminum dioxide. Optical simulation reveals that such a cooler has a high transparency for effective photons and strongly reflect waste photons. Besides, the thermal emissivity of the cooler is also very excellent. Thermal simulation results show that the temperature of a silicon solar panel can be passively reduced by over 5.7 °C by using the designed photonic cooler.

Similarly, other coolers were designed and investigated for the passive cooling of solar cells on the basis of the above-mentioned strategy. Slauch et al. [46] designed a 1D photonic mirror that is capable of reflecting a portion of waste photons while maintaining the transmission of effective photons, as shown in Fig. 10. Notably, the strong thermal emission of this structure is mainly due to the glass cover. Performance prediction indicates that the energy output of the PV system using this photonic mirror can be improved by 3.7% over a full year in Golden, Colorado, USA. In this case, the designed photonic cooler not only considers the spectra requirement but also considers the effect of the incident angle. Thus, the spectral property of the optimized cooler is not sensitive to the incident angle, and this is beneficial for for

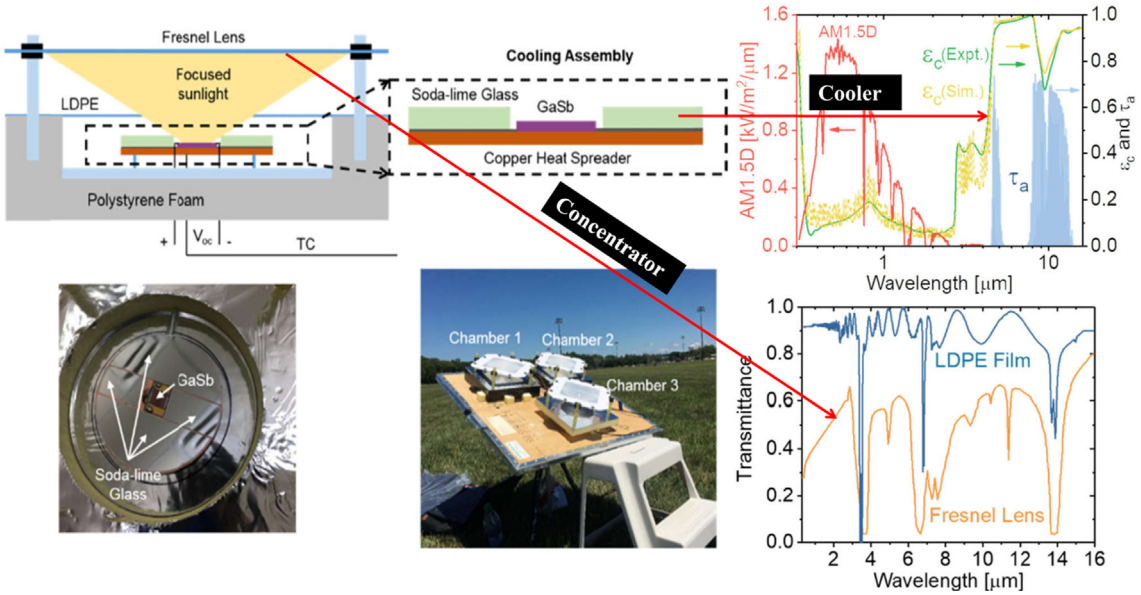


Fig. 8. Experimental setup of radiative cooling of solar cells in a CPV system. The spectral emissivity of the cooler (i.e., soda-lime glass) and the transmittance of the concentrator (i.e., Fresnel Lens) are presented as references. The figures and data are taken from Ref. [44].

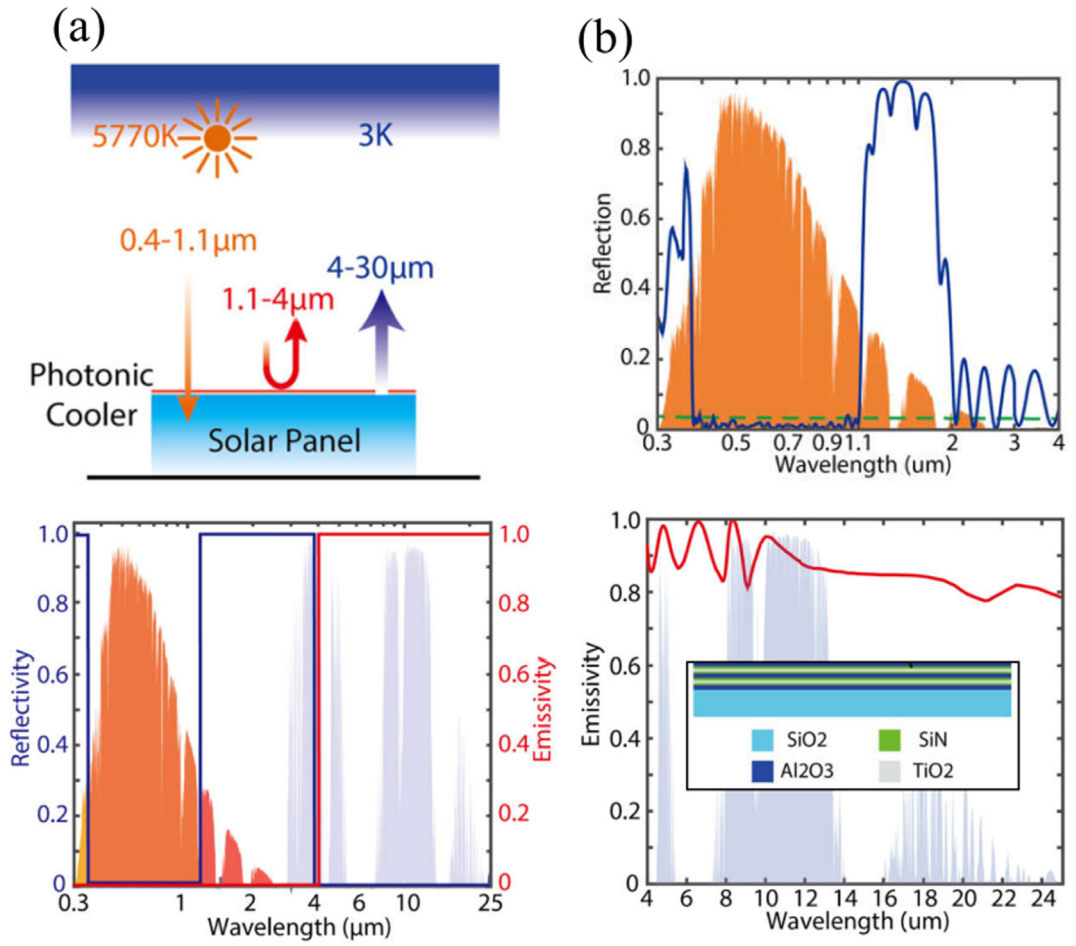


Fig. 9. (a) Schematic of full-spectrum thermal management of solar cells and the ideal spectral requirements of the cooler. (b) Calculated spectral properties of the designed photonic cooler with the structure of the designed cooler presented as a reference. The figures and data are taken from Ref. [25].

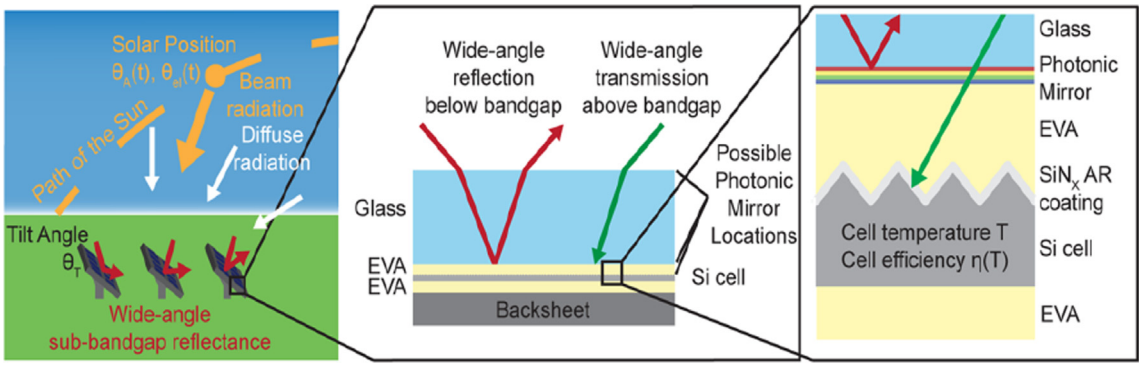


Fig. 10. Schematic of full-spectrum thermal management of PV modules on the basis of a photonic mirror. The figures are taken from Ref. [46].

real-world applications since the solar incident angle usually changes with time. Moreover, Zhao et al. [47] also designed a 2D photonic cooler to cool solar cells. The combined optical, thermal, and electrical simulation shows that the temperature of solar cells can be reduced approximately 8 °C over conventional PV panel. Analogously, An et al. [48] proposed a comprehensive multi-dimensional optical structure with a near-ideal spectral selectivity from sunlight to the infrared band for thermal management of solar cells. A temperature reduction of 10 °C and an absolute conversion efficiency improvement of 0.45% were predicted by a multi-physical opto-electro-thermal model.

To compare the effect of temperature reduction by using radiative cooling and full-spectrum thermal management, Sun et al. [42] conducted a comparative analysis to show the performance difference between these two approaches. As shown in Fig. 11, it is found that the temperature reduction of solar cells by using full-spectrum thermal management methods is obviously larger than that of radiative cooling, with a value of approximately 6 °C that is similar as the data reported in other literature [25], it proves that full-spectrum thermal management is a better approach to cool solar cells from the point of fundamental theory. Besides, Sun et al. [42] proposed a concept of using a spectrally selective mirror for low CPV application, which is a new attempt to control the temperature of the solar cells in the application. If the concentrator mirror is a selective mirror and the PV panel is equipped with a full-spectrum thermal management technology, the temperature of solar cells can be further reduced.

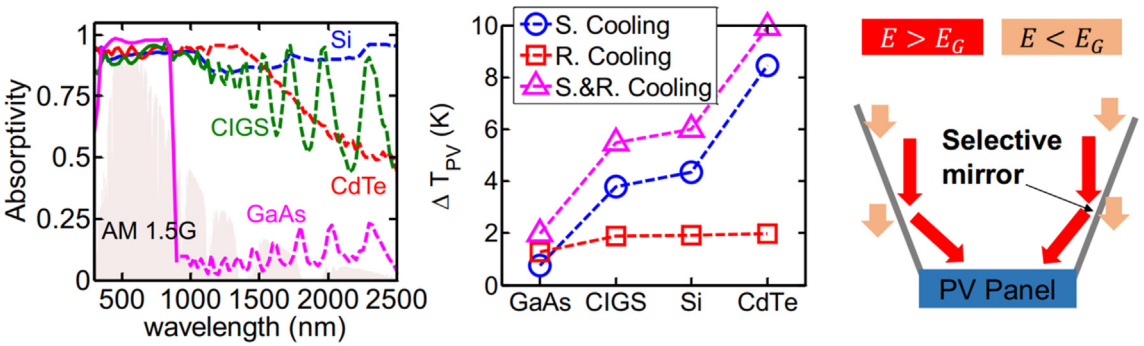


Fig. 11. Solar cells' temperature reduction comparison by using radiative cooling and full-spectrum thermal management methods involving different solar cells (left two figures), and the concept of using a selective mirror for low concentrated PV application. The figures and data are taken from Ref. [42].

3.3. Brief summary

Radiative cooling is a highly potential cooling technology for solar cells because radiative heat dissipation can successfully reduce the temperature of solar cells without the need for additional large components and extra energy inputs. However, in some pieces of literature related to radiative cooling of solar cells, the main purpose is to reveal this interesting phenomenon, thus the cooling effect of radiative cooling for solar cells may be overestimated. Specifically, commercial solar cells (e.g., highly doped silicon cell) and solar cells' encapsulation materials (e.g., glass) already have strong thermal emission, which indicates that these solar cells already possess radiative cooling ablity for self-cooling. Nevertheless, it is necessary to state that improving the thermal emission of encapsulated solar cells can still further contribute to solar cells' cooling, especially when the solar cells are used in certain special conditions, such as extraterrestrial photovoltaic application where is no air convection.

Compared with radiative cooling, using a spectrally selective filter to manage the sunlight and thermal emission of solar cells or PV modules is a more feasible and attractive method to cool solar cells. The solar cells' absorption of waste photons is one of the main heat sources for solar cells. For example, for silicon solar cells, the energy of waste photons is approximately 20% of the total solar radiation power, which will eventually be conveted into heat if absorbed by solar cells. The main component of the full-spectrum thermal management cooling technology is the spectrally selective filter. At present, the structure of the filter is relatively complex since the filter has completely different spectral requirements for multiple wavelength bands from ultraviolet to the mid-infrared band.

4. Challenges and prospects

At present, adopting spectrally selective approaches for passive cooling of solar cells is a research hotspot, which has drawn much attention from materials science to the engineering fields. However, there are some barriers to this cooling technology, which need to be considered for further development. Some issues, challenges, and prospects are discussed and presented as follows:

(1) A reliable and feasible cooler and filter are needed. Although there are numerous coolers or filters reported in the literature, the reliability is not sufficient. For example, silica is optically transparent and usually selected as one of the materials for cooler or filter design [32]. However, bulk silica is very fragile and cannot meet the mechanical stress/strain requirements as needed for industrial-scale photovoltaic applications. In the photovoltaic industrial, highly transparent glass has been widely used in photovoltaic panels [36], which has strong mechanical strength. Thus, spectrally selective filters can be designed and fabricated on the basis of photovoltaic glass in the future.

(2) The structure of the filter used in the full-spectrum thermal management approach is relatively too complex [25]. At present, a number of filters [25,47] are all optically designed and involve dozens of material layers, which indeed is difficult to fabricate. Besides, for the multilayer filter, it is necessary to evaluate the spectral property's sensitivity to the thickness of each layer since the thickness deviation from the designed value is unable to be avoided in the fabrication process [28]. Thus, a multilayer filter that has a suitable number of layers and is not sensitive to the thickness of layers will be the best design for real-world applications.

(3) The filter should be manufacturable at a large scale and at a low cost. At present, the fabrication of most reported filters requires equipment with micro and nanometer scale precision [25]. Thus, the fabricated filters will be very expensive and their scalability will be limited. However, it is necessary to state that if this filter is designed to cool solar cells under certain special cases, such as outer space [45], the above concern can be ignored.

(4) The optical property of the filter needs to be improved. The main feature of spectrally selective cooling approaches is to reduce the operating temperature of solar cells so that it can operate with higher electrical efficiency. Therefore, the effective solar absorption of solar cells should be maintained or improved after the filter is added so that optical losses do not exist to negate the aforementioned benefit [48].

(5) From the heat transfer point of view, an additional filter will increase the conductive thermal resistance between solar cells and ambient air [27], which will weaken the total cooling effect of solar cells after using the spectrally selective method. Thus, if the spectrally selective approach can be integrated directly into the solar cells' structure itself, the cooling potential of the spectrally selective method will be further enhanced.

5. Conclusions

In this paper, recent advancements in passive cooling of solar cells on the basis of spectrally selective approaches were reviewed, analyzed, discussed, and compiled. The energy balance of the solar cells and fundamental principles of the spectrally selective method for solar cells' cooling are introduced. Also, radiative cooling and full-spectrum thermal management approaches for passive cooling of solar cells are analyzed and discussed, which involve materials, optical design, and spectral property. Moreover, some challenges of this new technology were then discussed. Some significant conclusions are presented as follows:

(1) Radiative cooling of solar cells is a significant cooling method to control the temperature of solar cells. However, the encapsulated solar cells already have strong thermal emission, indicating that further improvements in the radiative cooling performance of solar

cells are limited. However, it is necessary to state that enhancing radiative cooling of solar cells is still a feasible way to cool solar cells in some cetain condiitons.

(2) Full-spectrum thermal management is a novel idea to cool solar cells passively by managing the light and thermal emission of the solar cell. The cooling effect of this method is obviously greater than that of radiative cooling.

(3) The filter used in spectrally selective approaches has a high requirement for materials. It's imperative that a reliable filter with the potential for larger-scale production within a feasible economic cost is found and/or fabricated, which is the key to this new cooling technology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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